

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458099.726 +/- 0.02 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.039
Transit depth (Rp/Rs)^2	1.89 +/- 1.08 [%]
Orbital Inclination (inc)	89.62 +/- 2.37
Airmass coefficient 1 (a1)	1.19 +/- 0.11
Airmass coefficient 2 (a2)	-0.102 +/- 0.021
Scatter in the residuals of the lightcurve fit is	9.3 %
Best Comparison Star	#1 - [361, 401]
Optimal Aperture	3.11
Optimal Annulus	10.85
Transit Duration (day)	0.09 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.138 ± 0.039 vs 0.1066 (deviation 0.53σ)
Duration fit vs published	0.09 d vs 0.1196 d (deviation 1.03σ)
Mid-time O-C	-30.2 min
Depth SNR	2.3
Residual scatter	9.3 %

Light curve

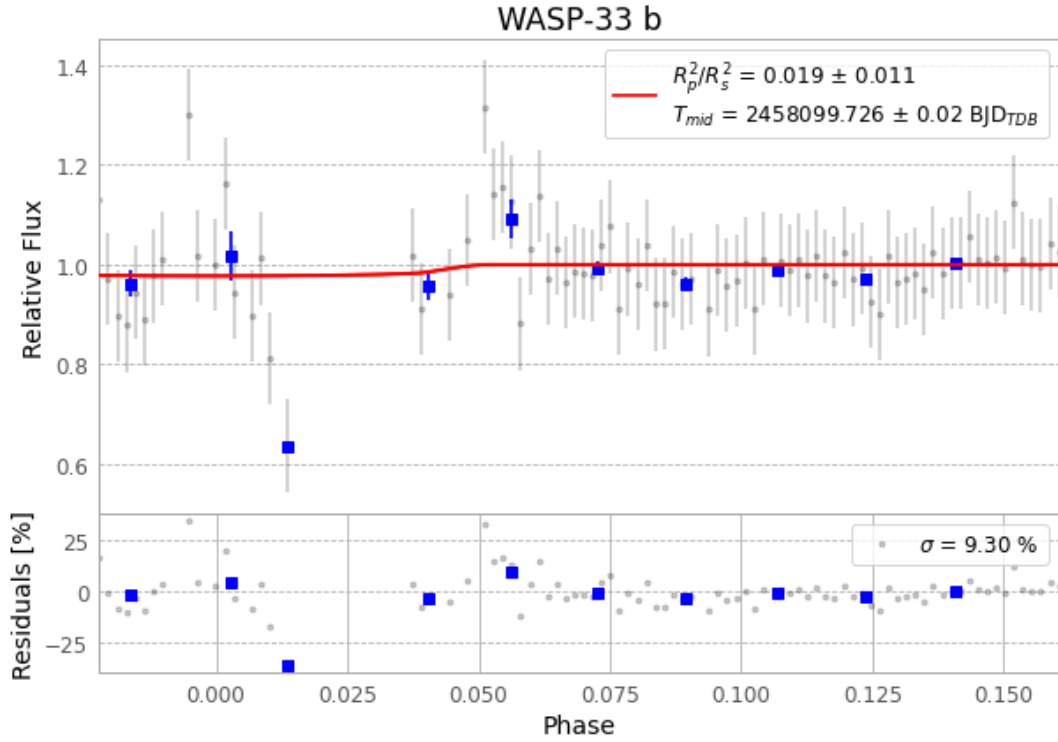


Figure 1 — FinalLightCurve_WASP-33 b_2017-12-11.png (SHA-256 f8efd2c2e6b9c48d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 301], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9db7b287bd1f8a33...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 96bb2ba

Data provenance

106 FITS frames (70 MB), WASP-33171212043912.FITS → WASP-33171212100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-33-171212.log (86213c248408...), FinalLightCurve_WASP-33 b_2017-12-11.png (f8efd2c2e6b9...), FinalLightCurve_WASP-33 b_2017-12-11.csv (af0a7d5fdbe0...)

Compute resources

Wall time 205.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 19:52Z.